

Year 10 Science
Sem. 1 Exam 2017

Format

Reading Time: 10 mins
Working Time: 100 mins

Structure

Section 1: 52 multiple choice
Section 2: 14 short answer
Section 3: Reading comprehension - 5 questions

Areas of Study

Astronomy

- Definitions.
- Distances in space - no light year calculations.
- Star lifecycles - stages and what happens in each.
- Stars - major element; process producing energy.
- Doppler Effect and red shift/blue shift - shifting of dark lines on spectrum.
- Hubble's observations of red shift - what did he measure?
- Origin of the universe - Big Bang versus Steady State; evidence; describe the Big Bang.
- Einstein's equation - $E = mc^2$ - relationship to matter being created in the early universe. No calculations involved.

Physical Science

- Calculations - $sp = \frac{d}{t}$ $a = \frac{v - u}{t}$ $v = u + at$ $F = ma$ $F_w = mg$
- Data sheet - will have the equations.
- Force - includes force forwards and friction acting backwards.
- Mass versus weight.
- Ticker timer tapes - no calculations.
- Newton's Three Laws - explanations and applications.
- Safety features of cars - crumple zones, air bags, etc. - how do they protect?
- Map - calculate distance; calculate distance and bearing.